

1.4 In-sample Calibration (ISC)

In contrast to EC, in-sample approach calibration (ISC) is characterized by an analytical calibration function obtained directly in each working sample. The SAM is probably the most established ISC procedure and popular in many fields, such as foods, environment, or forensic toxicology, where matrices are extremely variable, when the authentic analyte is available. Two other approaches also aim to simplify the quantification condition, depending on the chemical purity and the physicochemical properties of surrogate calibrant such as SIL. The former predicts the authentic analyte concentration by altering its natural isotopic pattern with a labeled analog standard. The latter is applicable when no significant interferences between the analyte and SIL are observed. In this case the authentic analyte concentration is directly determined.

1.4.1 Authentic Analyte: Standard Addition Method

As an operating procedure for absolute quantification, SAM consists in collecting the responses of authentic analyte additions in a series of aliquots obtained from the working sample. The simplest experimental design of SAM comprises a minimum of two runs described in Table 1.2. Notations are the same as in Figure 1.1:

- Level 0, or X_0 , is the *no-addition* level and consists in recording the response Y_0 in the working sample without any addition of the authentic analyte.
- Level 1, the working sample is spiked with a known amount of the authentic analyte.

By combining the two couples of data, the corrected concentration of the working sample is given by Equation (1.4).

Corrected concentration

$$Z = Y_0 \times \frac{X_1 - X_0}{Y_1 - Y_0} \quad (1.4)$$

The short worksheet below gives an example of computation. The formula applied in cell B5 is shown in cell C5. Figure 1.4 illustrates the data and shows that the corrected concentration corresponds to the extrapolation where Y-value is zero, and the line cuts the X-axis.

	A	B	C
1	Simple SAM		
2	Concentration X	Response Y	
3	0	32	
4	10	205	
5	Corrected result	1.8497	=B3*(A4-A3)/(B4-B3)

Table 1.2 Two-run experimental design of standard addition method.

	Concentration X	Response Y
Level 0 (no addition)	$X_0 = 0$	Y_0
Level 1 (spiked)	X_1	Y_1

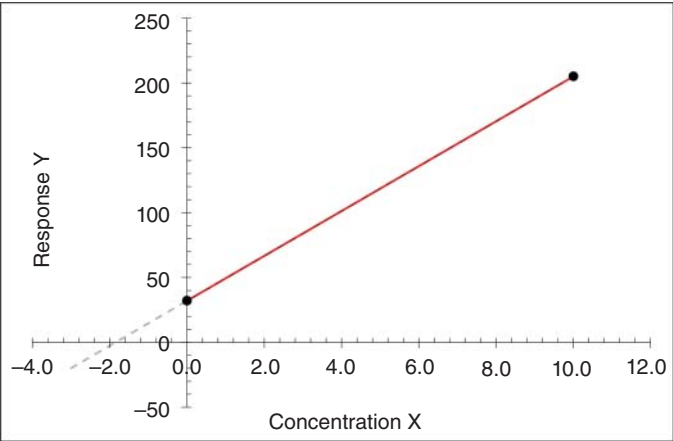


Figure 1.4 Two-run standard addition method.

This simplified experimental design can be routinely applied when each sample may have a specific matrix effect. For instance, when analyzing surface waters, it is classic to use simple SAM for each sample because the composition is recognized as highly variable. In this case, the result is obtained by combining two measurement values that are not replicated. A discussion about the role of replication in reducing MU is presented in Section 8.4.3.

Even simplified SAM is time-consuming with preparing and measuring two test portions per working sample. The benefit is to consider interindividual differences in matrix composition, to overcome matrix effects, and avoid building an EC curve. In that respect, it can be asserted as an absolute quantification method, as far as the response is exactly proportional to the concentration, in other words, linear.

As mentioned before, the FDA suggests applying SAM in a more complex experimental design to verify if using a surrogate matrix or analyte is justified. It calculates two calibration lines: one prepared by spiking several test portions of the working sample, the other by preparing calibrators with the surrogate matrix, which can be neat.

In the classic operating procedure, the working sample is divided into four and six identical aliquots, and a fortified calibration curve is obtained by spiking increasing known amounts of the authentic analyte, e.g. 50, 100, and 200% of the expected endogenous concentration. Only the first aliquot remains nonspiked, and

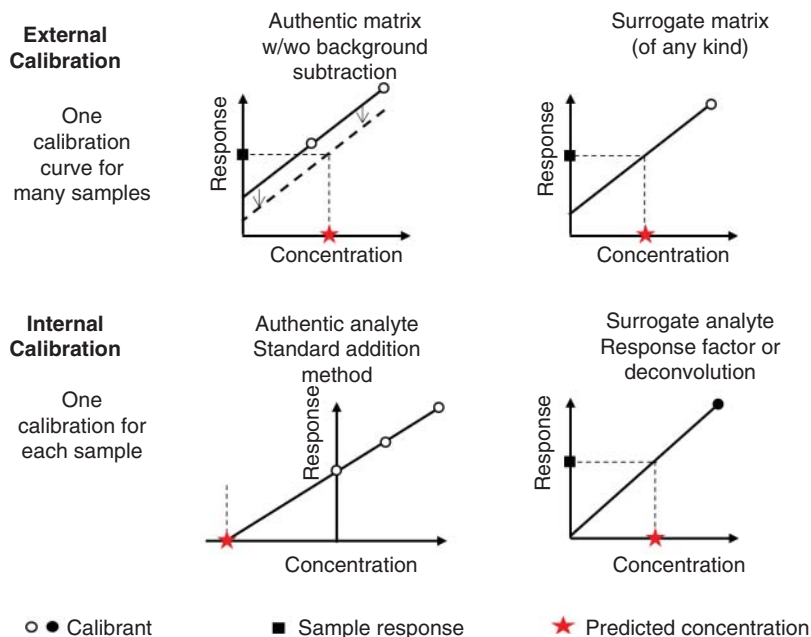


Figure 1.5 Calibration modes in analytical sciences. Source: Adapted from Visconti et al. [13].

its concentration is obtained by extrapolation where Y -value is equal to 0. This other protocol is illustrated with the example of Section 2.4.3. When the number of spikes is significant, SAM can also be applied when the calibration curve is polynomial, particularly when high endogenous signals affect the linearity of the response due to detector saturation.

When multiple signal-based detectors, such as MS or DAD, can record several physicochemical properties of the target analyte, more than one SAM calibration curve can be simultaneously acquired for the same working sample. This multiple-response monitoring leads to the possibility of dealing with the H-point standard addition method (HPSAM).

This new procedure is effective to control both proportional and additive biases (defined in Chapter 4), such as matrix interferences and/or detector saturation, when all calibration lines are converging at almost the same X -intercept. A comprehensive example is presented in Section 10.1. If the calibration lines are not correctly converging, a revised HPSAM was proposed including chemical modifiers [12]. Figure 1.5 is an attempt to propose a schematic overview of the diverse quantification/calibration strategies described in this chapter.

1.5 Some New Quantification Techniques

As stated, this chapter does not aim to give an exhaustive description of all possible quantification modes. However, it is valuable to describe some new insight on

a class of analytical techniques involving MS detection hyphenation, while many novel quantification modes were recently developed thanks to the improvement of modern MS instrumentation. More details are available in a recent review [13]. MS-coupled methods have progressively emerged as a one of the key instrumental components for numerous applications in laboratories, thanks to the development of new instruments and the reduction of costs.

The latter has become possible due to the advent of atmospheric pressure ionization interfaces, allowing to produce gas-phase ions that can be further analyzed. Compared to traditional spectroscopic detectors, such as UV absorbance, mass spectrometers offer additional selectivity by determining the mass/charge ratio of ion(s) or transition. An increasing number of articles reporting new MS-coupled methods for quantification are submitted each year [14].

In the field of MS-coupled methods, the greater availability of SILs opens the possibility of novel calibration procedures. They can mainly be employed as ideal surrogate calibrants to directly perform the calibration in the study matrix. Obviously, if they are used for this purpose, the analyst must first investigate the potential presence of interferences with the authentic analyte. When a contribution coming from the SIL is significant and modifies the signal, the application of isotope pattern deconvolution (IPD) was proposed as a corrective approach. In the absence of significant interference, internal calibration represents one of the most promising methodologies for modern absolute quantification.

1.5.1 Isotopic Pattern Deconvolution (IPD)

Isotope dilution mass spectrometry (IDMS) is a well-known technique applicable both to organic as inorganic analysis. It is because all isotopes of one element show almost the same chemical properties but mass differences between isotopes that IDMS allows quantifying the analyte by mass spectrometry. There are different IDMS operating procedures offering also various levels of precision. In many routine applications simple and fast operating procedures can be applied. The IPD is one of these high precision procedures based on the natural isotopic pattern alteration of a standard using a minor isotope labeled analog. In contrast to traditional analytical methods that rely on signal intensity, IPD is established by ratioing the signals between the isotopes of the molecule of interest and an analog with an enriched isotopic composition (i.e. SIL).

The IPD is sometimes claimed to be one of the most reliable and highest-quality metrological methods and is commonly used by chemical manufacturers to calculate SIL isotopic enrichment and purity. The isotopic abundance and concentration of the isotope labeled analog can be obtained by reverse isotope dilution mass spectrometry, i.e. a calibration against a high purity solution of the natural analyte prepared from a gravimetric solution of a suitable reference material.

First, the isotopic distributions for unlabeled standard and SIL as well as their combinations are computed using dedicated software: this is the convoluted isotope distribution. Free-access software is available coded with R to achieve the deconvolution. The labeled compound is then added to the reference material, resulting in

isotopic dilution. Then, the comparison between theoretical and experimental isotope overlap allows us to determine the SIL isotopic enrichment, chemical purity, and concentration.

Finally, once the SIL solution has been characterized by isotope dilution mass spectrometry it can be used as a calibrant for IPD quantification [15]. The more detailed procedural aspect is as follows, where variable A is the measured isotopic abundance, subscript nat comes for natural, lab for labeled and mix for mixed.

- Step 1. The natural isotopologue distributions of the analyte X and its isotope labeled analog X_{SIL} are measured. Let us remember that isotopologues only differ in their isotopic composition and have the same chemical formula. Superscripts $M0, M1$, etc. used in following formulas to indicate isotopologues.
- Step 2. Authentic analyte and SIL are mixed, and the resulting isotope pattern are determined. The basic concept is to say that the pattern of mixed solution is a linear combination of natural and labeled patterns weighted by the molar fractions q_{nat} and q_{lab} , respectively:

Deconvolution model for IPD

$$A_{mix} = q_{nat}A_{nat} + q_{lab}A_{lab} + E \quad (1.5)$$

The vector of random error E is added to account for the errors in the isotopic determinations. It is called a deconvolution model because it is slightly different of the classic calibration model, such as Equation (2.6), where there is only one predictive variable, the calibrant concentration usually noted X as explained in Sector 2.2. In this case there are two predictive variables A_{nat} and A_{lab} . Once the isotopic abundances are measured, we have a set of equations:

Isotopic patterns

$$\begin{aligned} A_{mix}^{M0} &= q_{nat}A_{nat}^{M0} + q_{lab}A_{lab}^{M0} + E^{M0} \\ A_{mix}^{M1} &= q_{nat}A_{nat}^{M1} + q_{lab}A_{lab}^{M1} + E^{M1} \\ A_{mix}^{M2} &= q_{nat}A_{nat}^{M2} + q_{lab}A_{lab}^{M2} + E^{M2} \\ &\dots \\ A_{mix}^{Mn} &= q_{nat}A_{nat}^{Mn} + q_{lab}A_{lab}^{Mn} + E^{Mn} \end{aligned}$$

They can be rewritten in a more condensed matrix form (the term matrix is used with its mathematical meaning) clearly showing this a multiple regression model with two variables and no intercept:

Multiple regression model

$$\mathbf{A}_{mix} = [\mathbf{A}_{nat} \mathbf{A}_{lab}] \mathbf{q}^{-1} + \mathbf{E} \quad (1.6)$$

- Step 3. Apply least-squares multiple linear regression to get the solutions of model 1.6; i.e. the estimates of the molar fractions q . With Excel this can be achieved using the `LINEST` built-in function. This function usage is described in Section 2.3.1. In this chapter, `LINEST` is applied to estimate the three coefficients of a quadratic model. Model in equation 1.6 is also a 3-coefficient model, with one coefficient equal to 0.